

The Geometry of Linear Algebra

Lecture 03 1: Row, Column, and Matrix Pictures

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October 22, 2025

These notes review the fundamental concepts for viewing a system of linear equations from three distinct but interconnected perspectives: the row picture, the column picture, and the matrix form. We will use a simple 2×2 system to illustrate how these geometric and algebraic interpretations provide a deeper understanding of the solution.

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1 Introduction and Central Example

The goal of this session is to solidify our understanding of the three primary ways to view a system of linear equations. To illustrate these concepts, we will work through a simple system of two equations with two unknowns, x and y .

Our central example for this lecture is:

$$\begin{cases} 2x + y = 3 \\ x - 2y = -1 \end{cases}$$

We will solve this system algebraically and then analyze its geometric meaning from the three different perspectives.

2 Solving the System Algebraically

Before exploring the geometry, let's find the unique solution using the method of substitution.

Step 1: Isolate a variable. From the second equation ($x - 2y = -1$), we can easily express x in terms of y :

$$x = 2y - 1$$

Step 2: Substitute. We substitute this expression for x into the first equation ($2x + y = 3$):

$$2(2y - 1) + y = 3$$

Step 3: Solve for y . Now, we simplify and solve the resulting single-variable equation:

$$4y - 2 + y = 3$$

$$5y - 2 = 3$$

$$5y = 5$$

$$y = 1$$

Step 4: Back-substitute to find x . With $y = 1$, we plug this value back into our expression for x :

$$x = 2(1) - 1$$

$$x = 1$$

Solution

The unique solution to the system is $(\mathbf{x}, \mathbf{y}) = (\mathbf{1}, \mathbf{1})$. This is the only pair of values that simultaneously satisfies both equations.

3 The Row Picture: Intersection of Lines

💡 Core Idea

The row picture considers each equation (or "row") in the system individually. In a 2D system, each linear equation represents a straight line on the xy -plane. The solution to the system is the single point where all the lines intersect.

3.1 Graphing the Lines

- **Line 1:** $2x + y = 3$
To draw this line, we find two points on it. If $x = 0$, then $y = 3$, giving point $(0, 3)$. If $x = 1$, then $y = 1$, giving point $(1, 1)$.
- **Line 2:** $x - 2y = -1$
Similarly, if $x = 0$, then $y = 1/2$, giving point $(0, 0.5)$. If $x = 1$, then $y = 1$, giving point $(1, 1)$.

3.2 Geometric Interpretation

When we plot both lines, we see they cross at a single point, which represents the solution to the system.

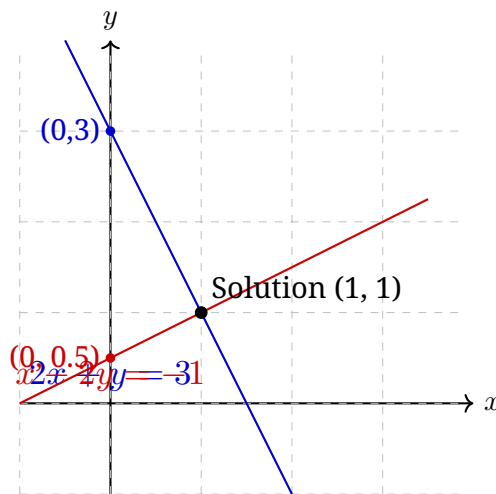


Figure 1: The Row Picture: Two lines intersecting at the solution point $(1, 1)$.

Key Takeaway

The row picture transforms the algebraic problem of solving a system into the geometric problem of finding the intersection point of two lines. The intersection is the solution.

4 The Column Picture: A Linear Combination of Vectors

Core Idea

The column picture reinterprets the system as a single vector equation. We group the coefficients of each variable into column vectors. The goal is to find the correct "linear combination" of these column vectors that produces the vector on the right-hand side.

4.1 Formulating the Vector Equation

We can rewrite the system by grouping terms by variable:

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

This gives us three key vectors:

- Column Vector 1: $\vec{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ (coefficients of x)
- Column Vector 2: $\vec{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ (coefficients of y)
- Target Vector: $\vec{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ (constants on the right side)

The question becomes: Find scalars x and y such that $x\vec{v}_1 + y\vec{v}_2 = \vec{b}$.

4.2 Geometric Interpretation

We know from our solution that $x = 1$ and $y = 1$. Let's visualize this combination:

$$1 \cdot \vec{v}_1 + 1 \cdot \vec{v}_2 = 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \vec{b}$$

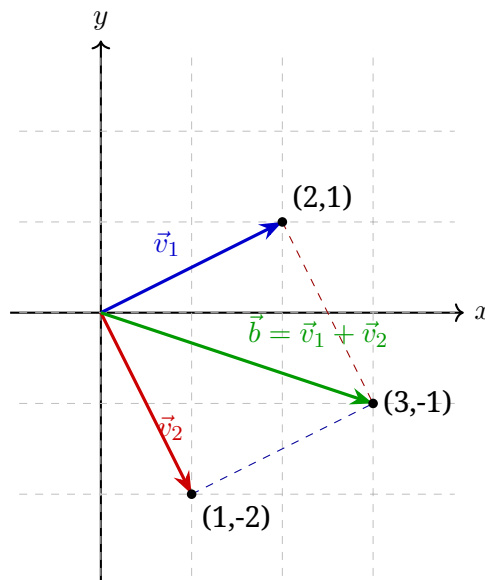


Figure 2: The Column Picture: The sum of vector $\vec{v}_1$ and vector $\vec{v}_2$ produces the target vector $\vec{b}$.

Key Takeaway

The column picture asks: how can we stretch/shrink the column vectors and add them together to reach the target vector? The solution (x, y) provides the exact scaling factors needed for this combination.

5 The Matrix Form: $A\mathbf{x} = \mathbf{b}$

Core Idea

The matrix form provides the most compact and powerful way to represent a linear system. It encapsulates all coefficient information into a single matrix A , the unknowns into a vector $\mathbf{x}$, and the constants into a vector $\mathbf{b}$.

5.1 Formulating the Matrix Equation

1. **Coefficient Matrix (A):** Formed by placing the column vectors side-by-side.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

2. **Vector of Unknowns ($\mathbf{x}$):**

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

3. **Constant Vector ($\mathbf{b}$):**

$$\mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

The entire system is now written as the compact equation $A\mathbf{x} = \mathbf{b}$:

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

Notice that the definition of matrix-vector multiplication, $A\mathbf{x}$, is precisely the linear combination of the columns of A shown in the column picture.

A Glimpse into Future Topics

Just as we can solve $ax = b$ with $x = a^{-1}b$, we can solve the matrix equation $A\mathbf{x} = \mathbf{b}$ using the concept of a **matrix inverse**, denoted A^{-1} . If it exists, the inverse has the property that $A^{-1}A = I$, where I is the identity matrix. The solution to the system can then be found directly as:

$$\mathbf{x} = A^{-1}\mathbf{b}$$

We will explore how to find A^{-1} and the conditions under which it exists later in the course. This concept is foundational for solving much larger and more complex systems.

6 Summary of the Three Perspectives

All three pictures describe the same problem but offer different insights. Being able to switch between them is a crucial skill in linear algebra.

Table 1: Comparison of the Three Pictures

Perspective	Geometric Interpretation	The Solution ($x=1$, $y=1$) Represents...
Row Picture	Two lines in a plane intersecting at a single point.	The coordinates of the intersection point.
Column Picture	The correct weighted sum (linear combination) of column vectors that produces a target vector.	The scalar weights needed for the linear combination of the column vectors.
Matrix Form	A linear transformation (by matrix A) of a vector (x) to another vector (b).	The input vector x that is mapped to the output vector b by the transformation A .